€ • , , ƒ 2
SCRATCH

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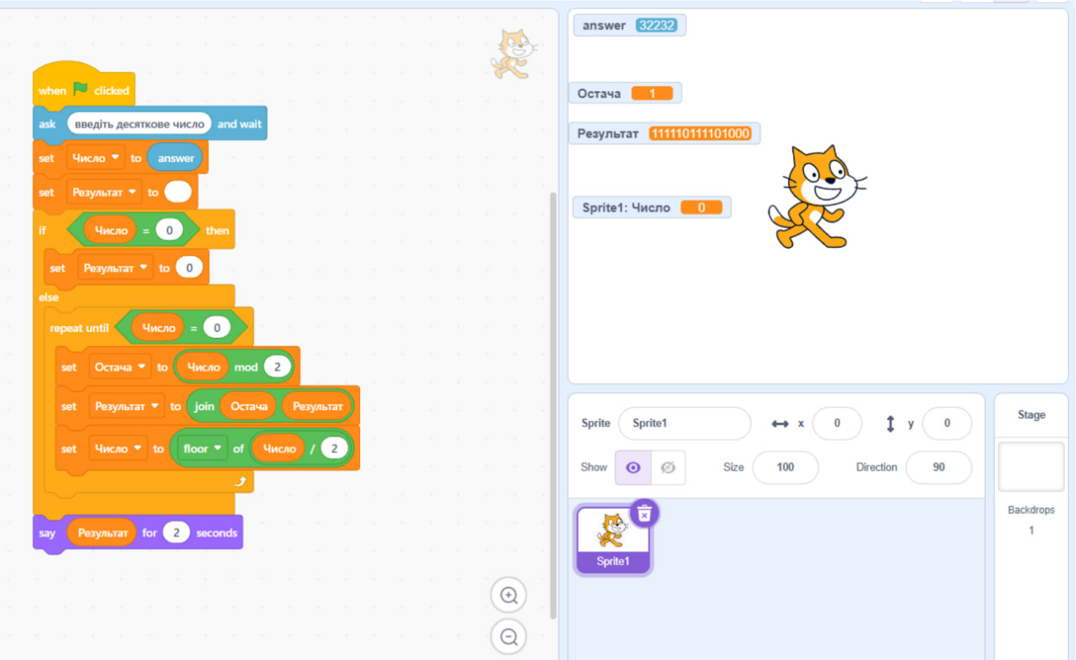
, : „ ˆ • •: , , , ‰ , , ˆ •
ˆ • ˆ ‰ Š ‹ .



```
when 🏴 clicked
ask введіть десяткове число and wait
set Число to answer
set Результат to ( )
if Число = 0 then
    set Результат to 0
else
    repeat until Число = 0
        set Остача to Число mod 2
        set Результат to join Остача Результат
        set Число to floor of Число / 2
say Результат for 2 seconds
```

answer 32232

Остача 1

Результат 111110111101000

Sprite1: Число 0

Sprite Sprite1, x 0, y 0, Show, Size 100, Direction 90

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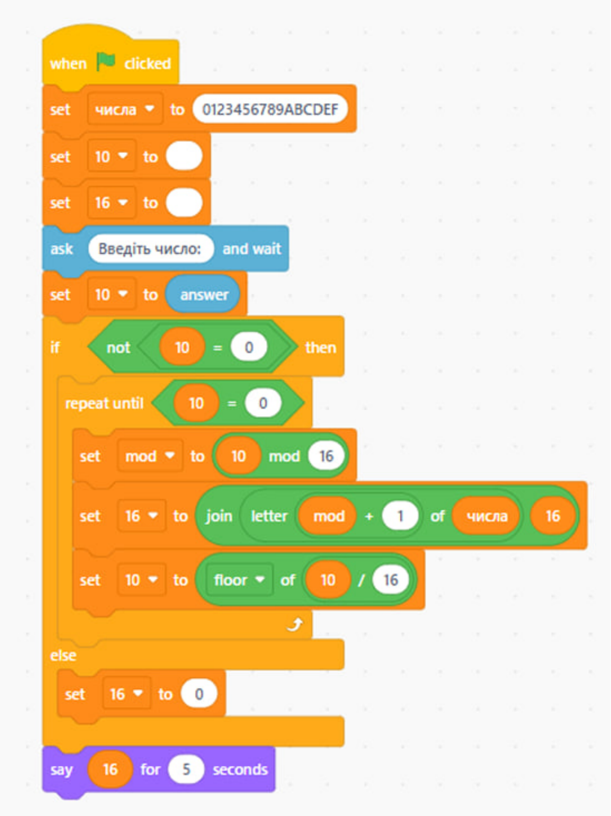
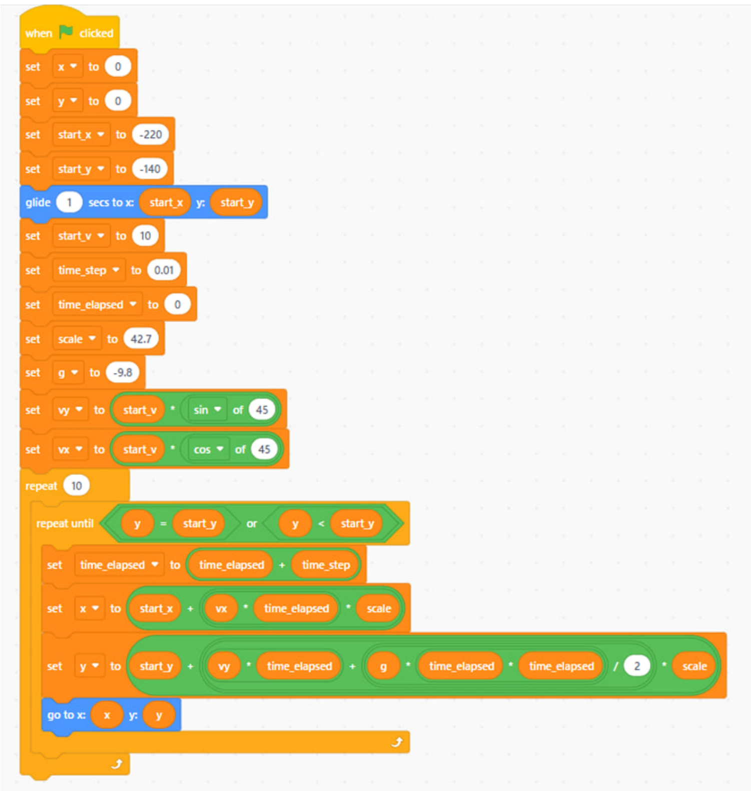
Š Scratch •‰ • ˆ • ‰ Š •
: ˆ •‰ ˆ • ˆ , ‰, ‰ Œ ˆ •
‰ Ž ‰ . • • 0, • ‰ ‰ 0 ‰ . „ ‹ ‰ ‰ Š •
, • Š • , ˆ 0. • ‘ •‰ ‘
ˆ • ˆ 2 (‰ Œ mod 2) ‰ ‰ ’ ˆ . “ •
ˆ ˆ ‘ • ‰ ‰ (join ’ ˆ Ž ‰),
• ˆ Š ˆ ‰ • . € • • ˆ ,
‰ ‹ ˆ ˆ • 2 (floor of Œ / 2). •
‰ • 2 .



```
when ⚑ clicked
set числа to 0123456789ABCDEF
set 10 to
set 16 to
ask Введіть число: and wait
set 10 to answer
if not 10 = 0 then
    repeat until 10 = 0
        set mod to 10 mod 16
        set 16 to join letter mod + 1 of числа 16
        set 10 to floor of 10 / 16
else
    set 16 to 0
say 16 for 5 seconds
```

€ • • ˆ ‹ •

Š Scratch- Š • ‰ • • ˆ ‹ • ‰ • . “ • • ‹ • ”0123456789ABCDEF• ‰ ˆ . ‡ Š : • ‰ ˆ • ‰ 10 , • • Š 0, ‰ •. • ‘ • ˆ • 16 (10 mod 16). ’ • ˆ ‘ , , ‹ ‰ 9, • • (‰ • ‰ +1 • Scratch), • , • ˆ ‰ . ’ • ˆ • 16, • ˆ , ˆ • ˆ 16 (floor of 10 / 16) ‘ . – Š •, ˆ • 0, ‹ • ˆ ‰ Š • 5 .



```
when [flag] clicked
set x to 0
set y to 0
set start_x to -220
set start_y to -140
glide 1 secs to x: (start_x) y: (start_y)
set start_v to 10
set time_step to 0.01
set time_elapsed to 0
set scale to 42.7
set g to -9.8
set vy to ((start_v) * (sin of 45))
set vx to ((start_v) * (cos of 45))
repeat 10
    repeat until <<(y) = (start_y)> or <(y) < (start_y)>>
        set time_elapsed to ((time_elapsed) + (time_step))
        set x to ((start_x) + (((vx) * (time_elapsed)) * (scale)))
        set y to ((start_y) + ((((vy) * (time_elapsed)) + ((((g) * ((time_elapsed) * (time_elapsed))) / 2))) * (scale)))
        go to x: (x) y: (y)
    end
end
```

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Š ‰ , ˆ , Š 45– ‰ . • ˆ ‰ • start_x = ˜220 start_y = ˜140, ˆ ‹ start_v = 10, ˆ time_step = 0.01, • g = ˜9.8 ‹ , scale = 42.7 • • . ™ ˆ sin cos , ˆ • Š ‹ vx vy.

Ž Š • ‰ : ‰ ‹ repeat 10 ‘ Š ‰ ˆ • Scratch, ‹ repeat until Š ˆ time_elapsed Š ‰ (x š , y š). … • Š • ˆ ‰ go to x: y:, ˆ , ‰ • start_y.

€ ˆ • , , • ˆ ˆ ‰ • Scratch. › : Š • • ˆ ‹ • ˆ •, ‘ • , 45– Š • ‘ • ‰ • ˆ . • , ˆ • ‰ ‰ ‰ , ˆ • ‰ ˆ • • .